

Research Article

Hybrid Models for Emotion Classification and Sentiment Analysis in Indonesian Language

Hendri Ahmadian ¹, Taufik F. Abidin ², Hammam Riza ³, and Kahlil Muchtar ⁴

¹Graduate School of Mathematics and Applied Sciences, Universitas Syiah Kuala, Banda Aceh 23111, Indonesia

²Department of Informatics, Universitas Syiah Kuala, Banda Aceh 23111, Indonesia

³Research Centre for Artificial Intelligence and Cyber Security, National Research and Innovation Agency, Jakarta 10340, Indonesia

⁴Department of Electrical and Computer Engineering, Universitas Syiah Kuala, Banda Aceh 23111, Indonesia

Correspondence should be addressed to Taufik F. Abidin; taufik.abidin@usk.ac.id

Received 27 December 2023; Revised 27 April 2024; Accepted 5 June 2024

Academic Editor: Anandakumar Haldorai

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The swift growth of social networks has enabled easy access to a wealth of user-created information for public assessment. These data hold potential for various uses, including analyzing comments and reviews through text analysis. The study utilizes a specialized version of the Bidirectional Encoder Representations from Transformers (BERT) model known as IndoBERT, tailored explicitly for Bahasa Indonesia. It aims to improve accuracy in the Indonesian natural language understanding benchmark by boosting performance in sentiment analysis and emotion classification tasks. The testing for both tasks involved a hybrid methodology that merged the summations of the four last hidden layers from the IndoBERT model with a combination of bidirectional long short-term memory (BiLSTM), bidirectional gated recurrent unit (BiGRU), and an attention model. The resulting model's performance was assessed using the F1-score metric. Based on the experimental results, the proposed model achieves an accuracy of 93% for sentiment analysis and 78% for emotion classification on the Indonesian natural language understanding (IndoNLU) benchmark dataset. The implementation result shows that the optimal accuracy of the models' performance evaluation was obtained using different hybrid models.

1. Introduction

Social media platforms have become indispensable in contemporary existence due to the rapid progress in information technology. The widespread use of these platforms has significantly altered modern society in numerous ways. Besides serving as venues for individuals to introduce themselves and share personal life narratives, these platforms also facilitate the expression of opinions and perspectives on a wide range of products, services, and organizational entities through posts and remarks [1]. As a result, users are generating an immense quantity of information. It becomes crucial for governments, organizations, and individuals to identify and harness substantial insights from this vast flow of user information [2, 3].

However, the widespread adoption and increasing significance of Internet technologies, especially social networking platforms, have presented formidable challenges in data management [4]. The continual growth in data volume poses difficulties in extracting valuable insights efficiently and effectively from these extensive datasets. In response to these challenges, scholars are actively exploring enormous data methodologies that can be implemented with enhancements in security, computational efficiency, storage, and access [5].

An additional obstacle surmounted was efficient data analysis, accomplished by developing an automated sentiment or opinion extraction technique. Sentiment analysis's primary advantage is its ability to accurately discern and categorize users' feelings and attitudes (positive, negative, or

neutral) about a specific product, subject, or service based on the textual content it produces. This analysis facilitates a more profound comprehension of an individual's perspective on a subject. [6].

The application of treatments has been instrumental in addressing emotion classification. This process autonomously assigns emotion categories to textual documents based on a predetermined set of categories. Given the escalating user base on virtual platforms and the proliferation of online content, assessing emotions conveyed in such material has garnered paramount importance for stakeholders, encompassing businesses, consumers, industry leaders, and other participants within the ecosystem [7].

The sentiment analysis process entails the automated extraction of sentiments, emotions, and opinions from texts [8]. Conversely, emotion classification pertains to accurately discerning and comprehending the feelings conveyed within a written scenario [9]. This task is frequently more challenging than sentiment analysis due to the more significant number of classes and subtle distinctions among them. While lexicon [10] and learning-based approaches [11, 12] have been utilized to address both types of instructions in the literature, a substantial quantity of labeled corpora is necessary to train the models accurately, which is frequently challenging and costly [13].

In recent years, substantial progress has been made in using pretrained transformer models in natural language processing (NLP). Transformers have proven effective in extracting textual sentences' semantic meaning [14] and syntactic structure [15]. However, it should be noted that this model predominantly prevails in languages with ample data sources, exemplified in the case of English. Scholars are increasingly adopting pretrained models based on transformers due to their proficiency in grasping universal linguistic representations. This approach obviates the need to train the model from its inception, which is advantageous in addressing diverse challenges within natural language processing [16]. To evaluate the proficiency of pretrained models in comprehending language contexts, foundational research on embeddings from language models [17] and Bidirectional Encoder Representations from Transformers [18] has played a pivotal role. Subsequently, general language models, such as Chinese Language Understanding and Evaluation [19] and General Language Understanding Evaluation [20] have been developed.

Creating the BERT architecture model depends on the datasets used for the Indonesian language. The twelve-task Indonesian natural language understanding (IndoNLU) benchmark (called IndoBERT) is the first task that needs to be done [21]. The present study employs hyperparameters derived from the BERT and ALBERT models [22]. The IndoBERT customizes the BERT architecture for the Indonesian language by compiling a large, clean corpus of Indonesian text, designing a specialized vocabulary and tokenizer, and training the model to recognize the unique linguistic features of the Indonesian language. Tackling morphological complexity, flexible

syntax, and semantic ambiguity challenges is critical for the model's effectiveness in understanding and processing Indonesian text [23].

Hybrid models have been devised by amalgamating BERT models with neural network models, tailored explicitly for addressing emotion classification and sentiment analysis tasks. The amalgamation of these two models has resulted in enhanced accuracy compared to utilizing either model in isolation. In addition, approaches incorporating BERT models have shown considerable proficiency in extracting vital features as feature vectors. Through this methodology, the input vectors for the neural network are constructed by aggregating features extracted from the final few hidden layers. This approach augments precision, particularly in emotion classification models and sentiment analysis.

Willie et al. [21] employed the BERT model for the Indonesian language, a.k.a. IndoBERT, for various NLP applications, focusing on emotion classification and sentiment analysis. They used the final hidden layer of BERT. They addressed the relatively low performance of the IndoBERT model in emotion classification and sentiment analysis tasks on the IndoNLU benchmark data. The model achieves an accuracy of 79% for emotion classification and 92% for sentiment analysis tasks. Consequently, there remains an opportunity to enhance the performance of the two NLP challenges. The improved IndoBERT model targeted the classification task aspect of the IndoBERT model to address the limitations of existing models by focusing on enhancing accuracy with a hybrid model architecture, thereby increasing performance.

Our study aims to refine the precision of the IndoBERT model in emotion classification and sentiment analysis tasks by using the IndoNLU benchmark dataset. The proposed approach introduces a hybrid model, combining the IndoBERT model with a neural network. This model features a customized IndoBERT layer [24] integrated with a neural network model. The primary innovation of this research lies in developing a hybrid model that combines the IndoBERT model with both BiLSTM and BiGRU models, explicitly tailored for optimal performance in emotion classification and sentiment analysis.

This paper's contributions include proposing hybrid models that combine the IndoBERT model with BiLSTM, BiGRU, and attention mechanism layer models for emotion classification and sentiment analysis tasks and evaluating their performance. We use the emotion classification (EmoT) dataset and sentiment analysis (SmSA) on the IndoNLU benchmark.

The remaining part of this paper has the following sections: The Section Related Works discusses the previous related works. The following section, Materials and Methods, explains the datasets used, methodologies, experimental setup, and parameter settings for the task. Section Results provide the outputs from the experiments and analysis. In the section Conclusions, we conclude with final remarks and future directions.

2. Related Works

The domains of sentiment analysis and emotion classification have recently emerged as significant research areas within natural language processing (NLP). Suciati and Budi [25] conducted a comparative study on machine learning and deep learning algorithms for sentiment analysis and emotion classification, employing a multilabel classification methodology. Ali et al. [26] utilized sentiment analysis and emotion classification models to rapidly detect pandemic outbreaks using machine learning and deep learning methodologies. Kakuba et al. [27] reviewed deep learning techniques' current trends and challenges for bimodal speech emotion recognition research. The authors also proposed deep learning for identifying and categorizing emotional reactions expressed in speech [28] and attention-based multi-learning for speech emotion recognition [29]. Many scholars have scrutinized the existing body of knowledge on sentiment analysis. Drus and Khalid [30] comprehensively examined sentiment analysis on social media through an in-depth literature study. The review delved into various methodologies, platforms, and applications. Additionally, Xu et al. [31] provided an exhaustive review of the challenges and potential impediments researchers encounter in performing sentiment analysis on social media.

Moreover, numerous investigations have been conducted in the domain of emotion classification. These investigations involve a comparative analysis of various transformer models and deep learning techniques. For instance, Lin and Nuha [32] and Glenn et al. [33] used the identical deep learning model for tasks involving the classification of emotions. Imaduddin et al. [34] utilized the BERT approach to do sentiment analysis on health application reviews. Lastly, extensive research efforts have been undertaken to tackle emotion categorization and sentiment analysis tasks, leveraging diverse techniques and expanding datasets to languages beyond English. The BERT model has been applied for emotion classification and sentiment analysis across various languages, including Chinese [35], Arabic [36], Italian [37], and Indonesian [38]. In Table 1, we summarize the reviewed studies.

3. Materials and Methods

In this study, we employ new hybrid models to enhance the accuracy of emotion and sentiment analysis tasks on the IndoBERT model of the IndoNLU benchmark. A hybrid approach was employed, combining the IndoBERT model (using the sum of the four final layers) with other neural network and attention mechanism layer models. Figure 1 depicts the combination. The deployed hybrid approach model includes

- (a) The IndoBERT model + attention
- (b) The IndoBERT model + bidirectional long short-term memory (BiLSTM) model
- (c) The IndoBERT model + BiGRU model

(d) The IndoBERT model + BiLSTM model + BiGRU model

(e) The IndoBERT model + BiGRU model + attention

The hybrid methodology combines these components to improve sentiment analysis and emotion classification tasks. Summing the outputs of the last four layers combines multiple levels of representation, capturing linguistic features. The BiLSTM and BiGRU models capture sequential dependencies in the feature representations extracted by IndoBERT. The attention mechanism allows the model to focus on specific parts of the input sequence that are most relevant for sentiment and emotion classification.

The illustration shown in Figure 1 can be elucidated in the following manner. Initially, the input, consisting of sentences, undergoes processing by the BERT tokenizer. Subsequently, the IndoBERT model utilizes the input ID attributes and attention masks the tokenizer provides to generate the embedding vector. Within the IndoBERT model, the process commences with the modifying IndoBERT architecture, explicitly employing the sum of the four last hidden layer methods. This modified input is then fed into the designated neural network model, which may entail the utilization of BiLSTM, BiGRU, or an attention model to enhance the comprehension of sentence semantics. Following this step, the outcomes of the preceding processes are forwarded to the neural network classifier, which incorporates a dropout layer to mitigate overfitting and a dense layer to procure results for each task. Ultimately, the model undergoes evaluation utilizing the confusion matrix technique.

Google Collaborative Environment is an assessment solution for running notebooks on Google Cloud Virtual Machines with hardware accelerators, notably Graphics Processing Units (GPUs), with 500 compute units. The IndoNLU benchmark dataset [39] was used to train, validate, and test the model in this study. It is a dataset specifically designed for emotion classification (EmoT) and sentiment analysis (SmSA). The EmoT dataset covers five emotion labels: anger, fear, happiness, love, and sadness.

Furthermore, the sentence-level sentiment analysis (SmSA) dataset consists of feedback and reviews in Indonesia were gathered from various Internet sources. This dataset was constructed by crawling the text and then categorizing it with the support of multiple Indonesian linguists. The SmSA dataset contains three labels: positive, negative, and neutral. Table 2 illustrates the description of the dataset.

4. Results

Table 3 outlines the parameters employed in the emotion classification and sentiment analysis tasks using a hybrid approach combining the IndoBERT and neural network models.

The F1-score was chosen as the primary metric to evaluate the suggested model's performance for sentiment analysis and emotion classification tasks because it strikes

TABLE 1: Deep learning-based emotion classification and sentiment analysis studies: overview.

Reference	Deep learning method	Aim of the study	Results
Lin and Nuha [32]	BERT, DistilBERT LSTM, and text convolutional network	To improve the efficiency of sentiment analysis based on textual data	The hybrid model that was suggested exhibited superior performance in sentiment classification metrics
Glenn et al. [33]	BiLSTM	To perform emotion classification of Indonesian tweets using BiLSTM models	BiLSTM models achieved more than an 8% gain in accuracy compared to traditional machine learning techniques
Imaduddin et al. [34]	The BERT model for Indonesian language (IndoBERT)	To create a sentiment analysis model using reviews from health service applications on the Google Play Store	The study achieves impressive results with an F1-score of 95%
Li et al. [35]	BERT, recursive convolutional neural network, attention mechanism	To classify users' emotions based on BERT, recursive convolutional neural network (RCNN), and attention mechanism model	It attains a high level of precision in getting contextual information, enhances accuracy in classification
Abdelgwad et al. [36]	BERT	Performed extensive research on three Arabic datasets, using the BERT embedding component with a basic linear classification layer	The experimental results demonstrate that it outperformed the most advanced work and is resistant to overfitting
Chiorrini et al. [37]	BERT-based language understanding model for the Italian language (ALBERTo)	Performed training and evaluation on two different Italian Twitter datasets to evaluate their efficacy in recognizing emotion	The models reveal exceptional performance in both 4-class and 6-class emotion classifications
Setiawan et al. [38]	BERT	Created and evaluated the Indonesian dataset for emotion analysis from Facebook and ISEAR Indonesian translated dataset	Achieved the highest accuracy of 79% with the fine-tuning BERT model

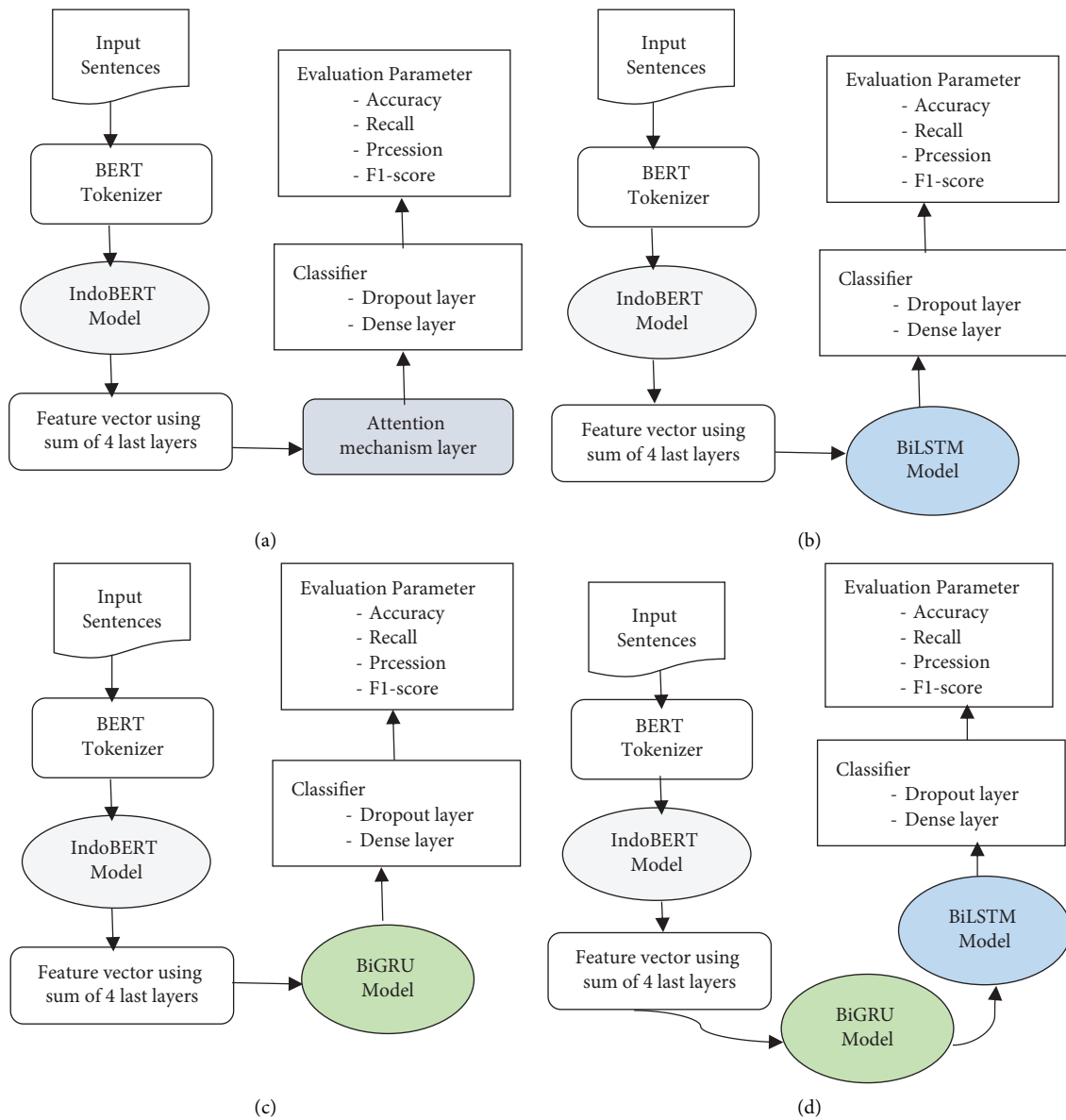


FIGURE 1: Continued.

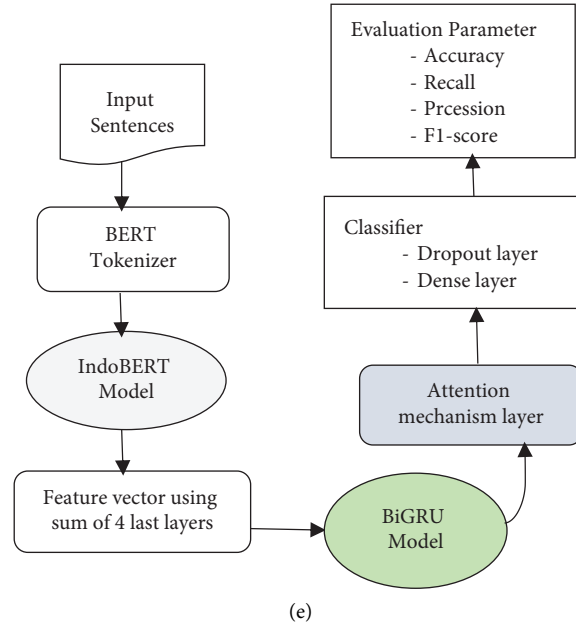


FIGURE 1: Hybrid model architecture. (a) The IndoBERT model (sum of 4 last layers) + attention. (b) The IndoBERT model (sum of 4 last layers) + BiLSTM. (c) The IndoBERT model (sum of 4 last layers) + BiGRU. (d) The IndoBERT model (sum of 4 last layers) + BiLSTM + BiGRU. (e) The IndoBERT model (sum of 4 last layers) + BiGRU + attention layer.

TABLE 2: Summary of datasets.

No	Dataset	Training	Validation	Testing	Task description
1	EmoT	3.521	440	442	Emotion classification
2	SmSA	11.000	1.260	500	Sentiment analysis

TABLE 3: Optimal hyperparameters of the hybrid model.

Hyperparameter	Value
Epoch	10
Batch size	16
Max sequence	512
Learning rate	0.000003
Optimizer	Adam
Activation function	Softmax
Loss function	Categorical cross-entropy
Model pretrained	Base phase 1, base phase 2, large phase 1, and large phase 2

a balance between precision and recall, making it a robust measure of model effectiveness. The F1-score, represented by

equation (3), is calculated as the harmonic mean of precision and recall. As indicated by equation (1), precision quantifies the degree to which the system reliably identifies the pertinent entities out of all the entities it has recognized. Precision is quantified by the ratio of accurately recognized entities to the total number of entities detected by the system. As seen in equation (2), recall quantifies the system's ability to recognize a significant number of relevant entities from the total number of entities in the data. Recall is defined as the quotient of the number of correctly identified entities, the sum of the number of correctly identified entities, and the number of unidentified entities detected by the system. Lastly, accuracy, represented by equation (4), measures the number of correct samples to all samples. Higher accuracy indicates a better performance.

$$\text{Accuracy} = \frac{\text{TruePositive} + \text{TrueNegative}}{\text{TruePositive} + \text{TrueNegative} + \text{FalsePositive} + \text{FalseNegative}}, \quad (1)$$

$$\text{Precision} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}}, \quad (2)$$

$$\text{Recall} = \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}}, \quad (3)$$

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \quad (4)$$

4.1. Results of Model Hybrid Testing in the EmoT Dataset. Table 4 summarizes the test results of applying the hybrid model to the emotion classification tasks using the EmoT dataset.

Table 4 outlines crucial features of the hybrid model that combines the IndoBERT and neural network models. The mixed model technique does not enhance accuracy in pretrained base or large models. The hybrid model, which combines the IndoBERT-modified layer model with the BiLSTM model, achieves a marginal increase of 0.01, reaching a value of 0.78 in large phase 1. Conversely, the hybrid model that combines the IndoBERT model with the BiGRU model gets a value of 0.76 in base phase 1. However, the accuracy value remained unchanged in the base and large hybrid models, including the IndoBERT layer modification and the BiGRU models. The accuracy value does not improve in the model that combines the IndoBERT layer modification model and attention layer, the layer modification and BiLSTM models plus BiGRU model, and finally, the layer modification and BiGRU models plus attention layer.

We discovered that the hybrid models' performance is generally lower than the original IndoBERT model on the EmoT dataset. These results are consistent with the work done in [40], which applied the BERT method on the ISEAR dataset with an F1-score value of 73%.

4.2. Training and Validation Loss and Accuracy of the EmoT Dataset. In this subsection, the training and validation loss and accuracy outputs for each hybrid model are selected only for the pretrained base phase 2 and large phase 2 models because they have outputs similar to phase 1.

4.2.1. The IndoBERT + BiLSTM Models. The hybrid model's training and validation loss and accuracy numbers indicate that the training loss lowers while the validation loss begins to climb by epoch 3. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains constant from epoch 3 to 10 in both the pretrained base and large models. Figures 2(a) and 2(b) illustrate the training and validation loss, and Figures 3(a) and 3(b) show training and validation accuracy.

4.2.2. The IndoBERT + BiGRU Models. The results of this hybrid model show that training loss decreases and validation loss increases at epoch 4. While training accuracy increases until epoch 10, validation loss offers a constant value from epoch 3 to 10, both on pretrained base and large model. An illustration of training and validation loss is

shown in Figures 2(c) and 2(d), and that of training and validation accuracy is shown in Figures 3(c) and 3(d).

4.2.3. The IndoBERT Model + Attention. The hybrid model's training and validation loss and accuracy indicate that the training loss lowers, but the validation loss grows starting from epoch 6. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains constant from epoch 2 to 10 in both the pretrained base and large models. Figures 2(e) and 2(f) visually represent the training and validation loss, and Figures 3(e) and 3(f) show training and validation accuracy.

4.2.4. The IndoBERT + BiLSTM + BiGRU Models. The hybrid model's training and validation loss and accuracy information indicate that the training loss lowers, but the validation loss rises starting from epoch 6. The training accuracy steadily improves until epoch 10, whereas the validation loss remains constant from epoch 3 to 10 for both the pretrained base and large models. Figures 2(g) and 2(h) depict the training and validation loss, and Figures 3(g) and 3(h) show training and validation accuracy.

4.2.5. The IndoBERT + BiGRU + Attention Models. This hybrid model's training and validation loss and accuracy findings indicate that the training loss decreases, but the validation loss begins to grow starting from epoch 5. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains constant from epoch 2 to epoch 10. This observation is consistent with using a pretrained base and a large model. Figures 2(i) and 2(j) show the visual representation of the training and validation loss, and Figures 3(i) and 3(j) show training and validation accuracy.

4.3. Results of Model Hybrid Testing in the SmSA Dataset. Table 5 presents a concise overview of the evaluation findings achieved by utilizing the hybrid model for sentiment analysis tasks using the SmSA dataset. Table 5 yields significant findings in the hybrid of the IndoBERT and neural network models. Overall, the hybrid model method enhances accuracy in both pretrained base and large models. In the IndoBERT model, precisely the sum of the last four layers, the BiLSTM model performance increase of 0.02 was observed in base phase 1. In addition, a 0.90 increase was observed in phase 2. In large phase 1, more than 0.01 increase was observed, resulting in a value of 0.93.

Furthermore, during the pretrained base phase 2, the accuracy increased by 0.90. The IndoBERT model, consisting of the sum of the last four layers, combined with the BiGRU

TABLE 4: The hybrid model evaluation results using the EmoT dataset for emotion classification tasks.

Hybrid model	Model pretrained	Accuracy	Recall	Precision	F1-score
IndoBERT + BiLSTM	Base phase 1	0.75	0.75	0.76	0.76
	Base phase 2	0.74	0.75	0.76	0.75
	Large phase 1	0.78	0.78	0.77	0.78
	Large phase 2	0.73	0.74	0.74	0.74
IndoBERT + BiGRU	Base phase 1	0.75	0.75	0.77	0.76
	Base phase 2	0.75	0.76	0.75	0.75
	Large phase 1	0.75	0.76	0.76	0.76
	Large phase 2	0.75	0.76	0.75	0.75
IndoBERT + BiLSTM + BiGRU	Base phase 1	0.72	0.72	0.74	0.73
	Base phase 2	0.73	0.74	0.75	0.74
	Large phase 1	0.74	0.76	0.75	0.75
	Large phase 2	0.72	0.73	0.73	0.73
IndoBERT + attention layer	Base phase 1	0.72	0.73	0.72	0.72
	Base phase 2	0.72	0.73	0.72	0.73
	Large phase 1	0.77	0.77	0.78	0.77
	Large phase 2	0.76	0.77	0.78	0.77
IndoBERT + BiGRU + attention layer	Base phase 1	0.74	0.74	0.74	0.74
	Base phase 2	0.74	0.75	0.75	0.75
	Large phase 1	0.76	0.76	0.77	0.77
	Large phase 2	0.76	0.77	0.78	0.77
IndoBERT [39]	Base phase 1				0.75
	Base phase 2				0.76
	Large phase 1				0.77
	Large phase 2				0.79

The bold values indicate that the performance is better than the baseline model (IndoBERT's performance).

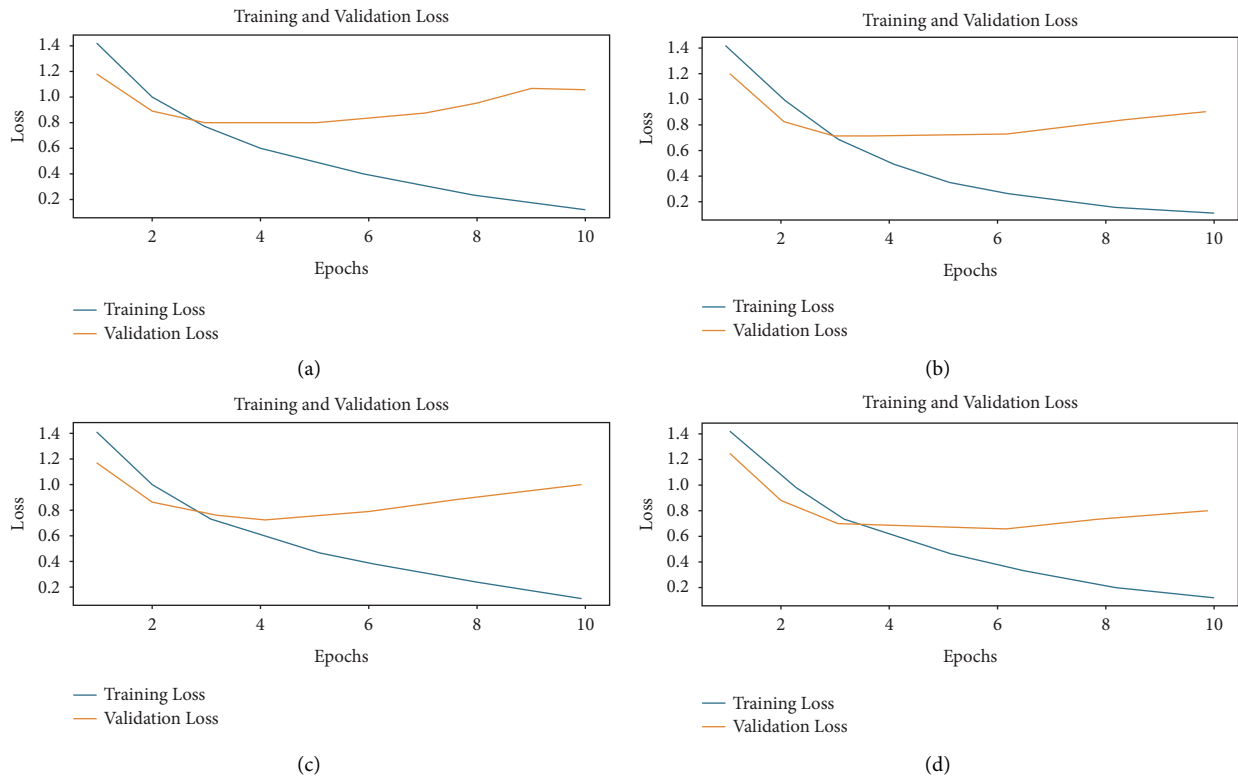


FIGURE 2: Continued.

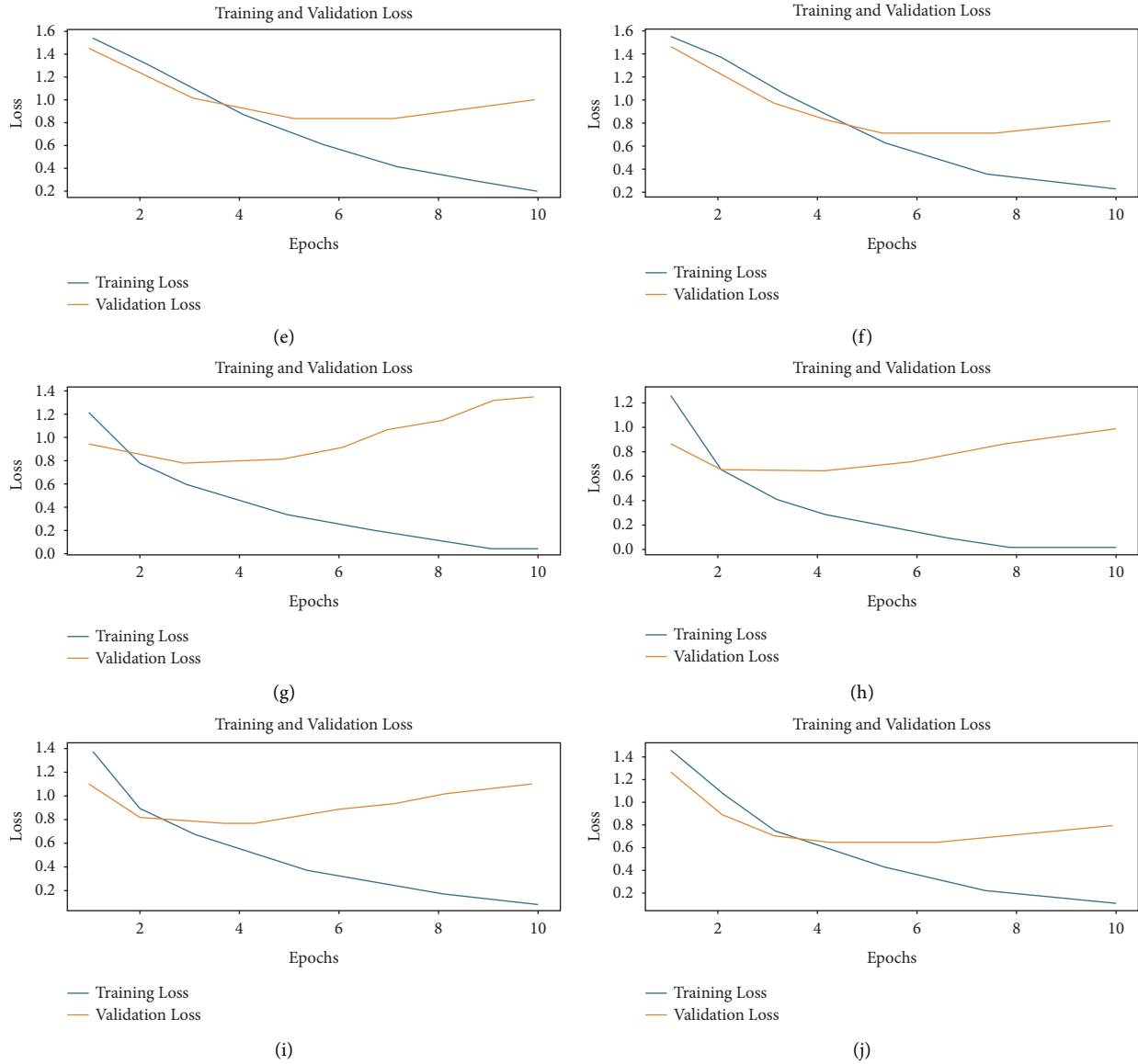


FIGURE 2: The illustration of training and validation loss for all models of the emotion classification tasks. (a) The output of the training and validation loss of the IndoBERT and BiLSTM models in the pretrained base. (b) The output of the training and validation loss of the IndoBERT and BiLSTM models in the pretrained large. (c) The output of the training and validation loss of the IndoBERT and BiGRU models in the pretrained base. (d) The output of the training and validation loss of the IndoBERT and BiGRU models in the pretrained large. (e) The output of the training and validation loss of the IndoBERT, BiLSTM, and BiGRU models in the pretrained base. (f) The output of the training and validation loss of the IndoBERT, BiLSTM, and BiGRU models in the pretrained large. (g) The output of the training and validation loss of the IndoBERT and attention models in the pretrained base. (h) The output of the training and validation loss of the IndoBERT and attention models in the pretrained large. (i) The output of the training and validation loss of the IndoBERT, BiGRU, and attention models in the pretrained base. (j) The output of the training and validation loss of the IndoBERT, BiGRU, and attention models in the pretrained large.

model and the attention layer, demonstrated a significant improvement in accuracy. In base phase 1, the accuracy increased by 0.88. In base phase 2, it reached an accuracy of 0.89.

In addition, the IndoBERT model, which consists of the sum of the last four layers, combined with the BiLSTM and BiGRU models, achieved an accuracy of 0.91 during the pretrained base phase 1. Furthermore, during the pretrained base phase 2, the accuracy increased by 0.90. The IndoBERT

model, consisting of the sum of the last 4 layers, combined with the BiGRU model and the attention layer, demonstrated a significant improvement in accuracy. In base phase 1, the accuracy increased by 0.88, and in base phase 2, it reached an accuracy value of 0.89.

In general, the experimental results on the SmSA dataset are compared to the accuracy of the original IndoBERT model. The hybrid model demonstrates a 93% higher accuracy than the original IndoBERT model.

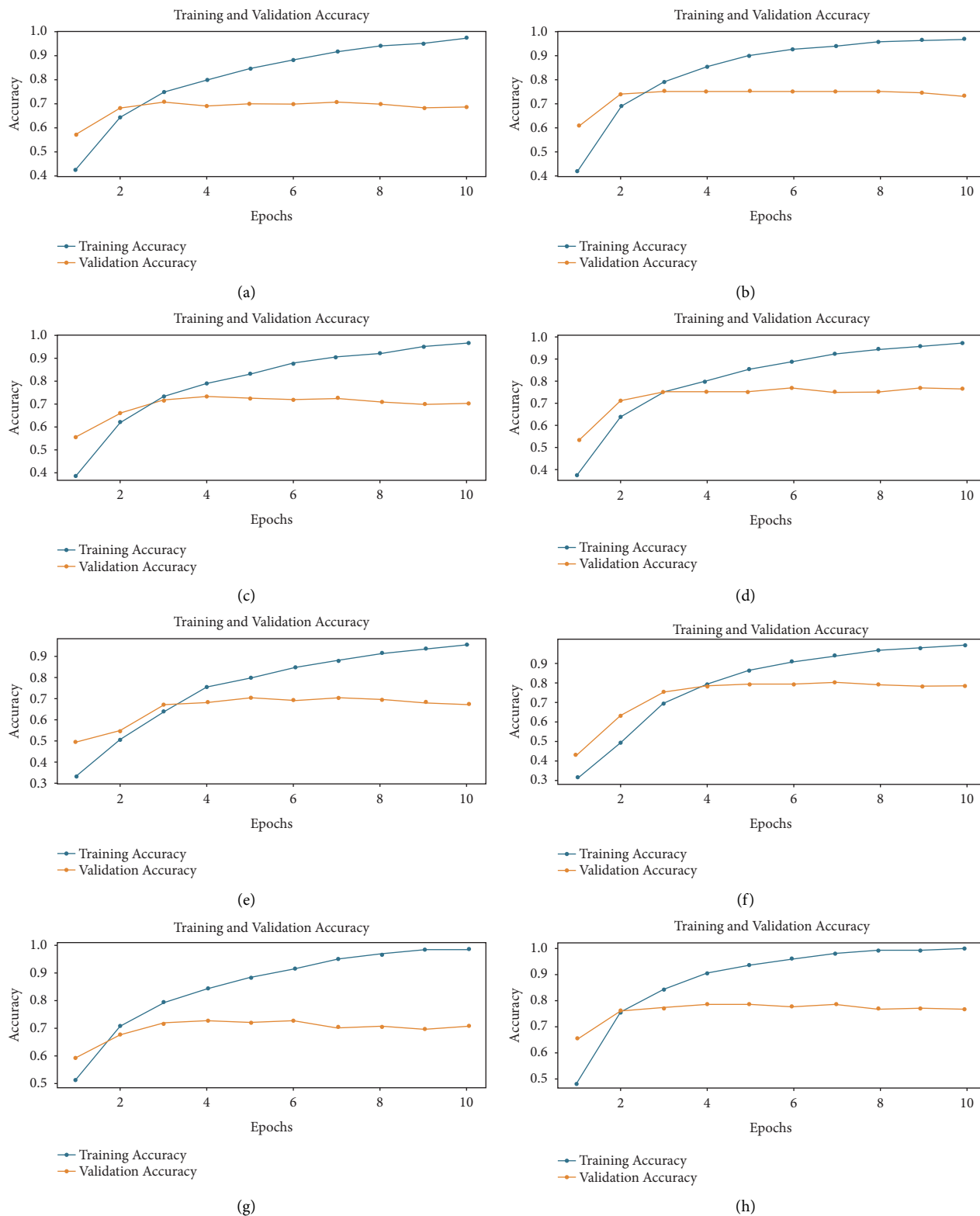


FIGURE 3: Continued.



FIGURE 3: The illustration of training and validation accuracy for all models of the emotion classification tasks. (a) The output of the training and validation accuracy of the IndoBERT and BiLSTM models in the pretrained base. (b) The output of the training and validation accuracy of the IndoBERT and BiLSTM models in the pretrained large. (c) The output of the training and validation accuracy of the IndoBERT and BiGRU models in the pretrained base. (d) The output of the training and validation accuracy of the IndoBERT and BiGRU models in the pretrained large. (e) The output of the training and validation accuracy of the IndoBERT, BiLSTM, and BiGRU models in the pre-trained base. (f) The output of the training and validation accuracy of the IndoBERT, BiLSTM, and BiGRU models in the pre-trained large. (g) The output of the training and validation accuracy of the IndoBERT and attention models in the pre-trained base. (h) The output of the training and validation accuracy of the IndoBERT and attention models in the pretrained large. (i) The output of the training and validation accuracy of the IndoBERT, BiGRU, and attention models in the pretrained base. (j) The output of the training and validation accuracy of the IndoBERT, BiGRU, and attention models in the pretrained large.

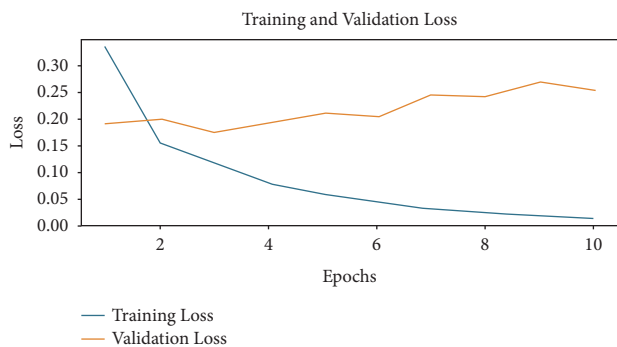
TABLE 5: Hybrid model evaluation results using the SmSA dataset for sentiment analysis tasks.

Hybrid model	Model pretrained	Accuracy	Recall	Precision	F1-score
IndoBERT + BiLSTM	Base phase 1	0.92	0.88	0.93	0.90
	Base phase 2	0.92	0.88	0.92	0.90
	Large phase 1	0.94	0.92	0.94	0.93
	Large phase 2	0.92	0.86	0.93	0.88
IndoBERT + BiGRU	Base phase 1	0.91	0.85	0.92	0.87
	Base phase 2	0.92	0.88	0.92	0.89
	Large phase 1	0.94	0.90	0.94	0.92
	Large phase 2	0.93	0.89	0.94	0.91
IndoBERT + BiLSTM + BiGRU	Base phase 1	0.93	0.90	0.92	0.91
	Base phase 2	0.92	0.88	0.92	0.90
	Large phase 1	0.93	0.91	0.93	0.92
	Large phase 2	0.93	0.90	0.93	0.91
IndoBERT + attention layer	Base phase 1	0.91	0.86	0.92	0.88
	Base phase 2	0.91	0.85	0.92	0.87
	Large phase 1	0.93	0.88	0.94	0.90
	Large phase 2	0.93	0.90	0.94	0.91
IndoBERT + BiGRU + attention layer	Base phase 1	0.91	0.86	0.93	0.88
	Base phase 2	0.91	0.86	0.91	0.89
	Large phase 1	0.93	0.90	0.93	0.91
	Large phase 2	0.94	0.90	0.94	0.92
IndoBERT [39]	Base phase 1				0.87
	Base phase 2				0.87
	Large phase 1				0.92
	Large phase 2				0.92

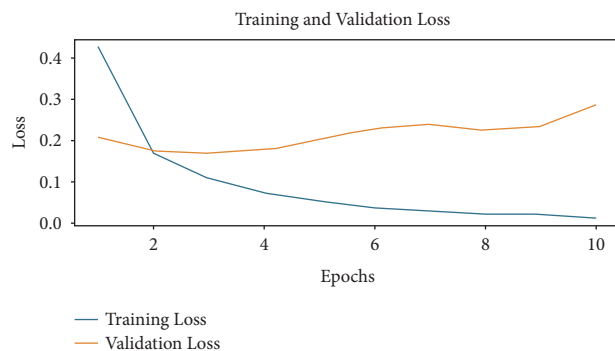
The bold values indicate that the performance is better than the baseline model (IndoBERT's performance).

4.4. Training and Validation Loss and Accuracy of the SmSA Dataset. This part only looks at the outputs of training and validation loss and training and validation accuracy for each hybrid model for the base and large phase 2 models in the analysis sentiment tasks because their outputs are the same as phase 1.

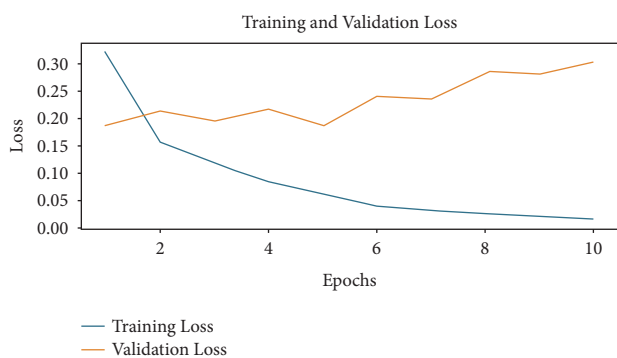
4.4.1. The IndoBERT + BiLSTM Models. The hybrid model's training and validation loss and accuracy indicate a decrease in training loss and a simultaneous increase in validation loss starting from epoch 6. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains constant from epoch 2 to 10 in both



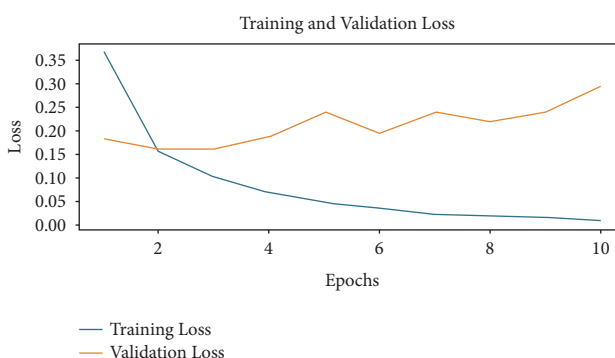
(a)



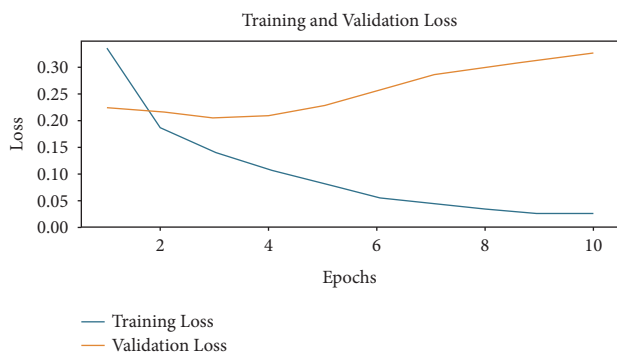
(b)



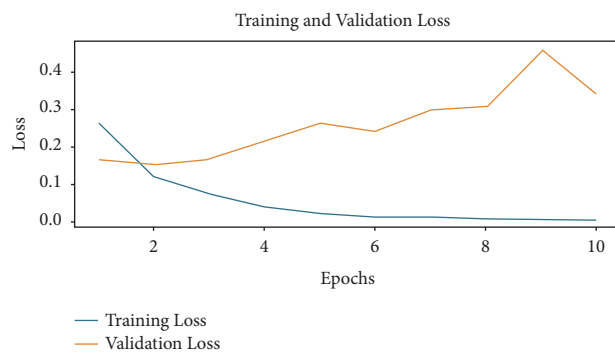
(c)



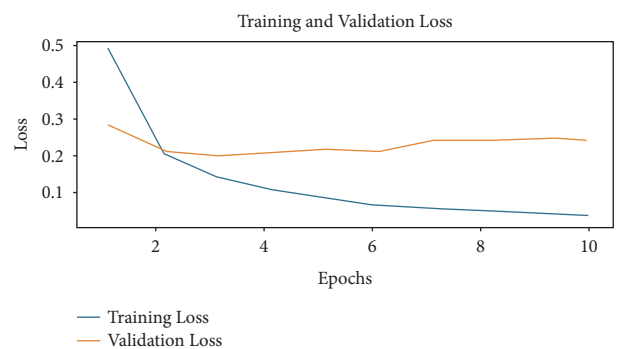
(d)



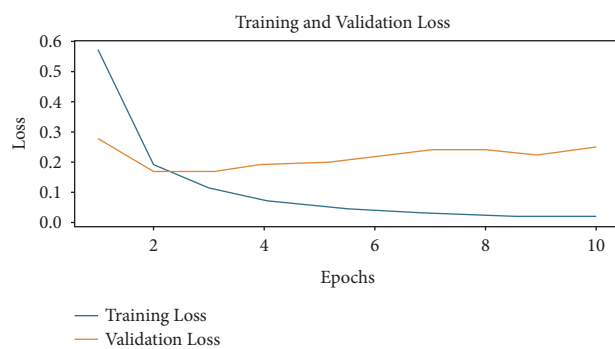
(e)



(f)



(g)



(h)

FIGURE 4: Continued.

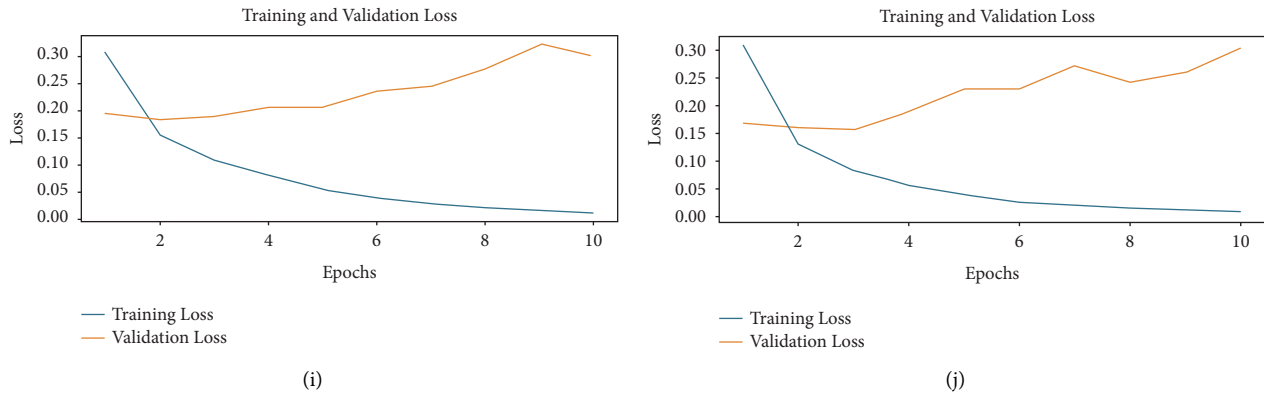


FIGURE 4: The illustration of training and validation loss for all models of the sentiment analysis tasks. (a) The output of the training and validation loss of the IndoBERT and BiLSTM models in the pretrained base. (b) The output of the training and validation loss of the IndoBERT and BiLSTM models in the pretrained large. (c) The output of the training and validation loss of the IndoBERT and BiGRU models in the pretrained base. (d) The output of the training and validation loss of the IndoBERT and BiGRU models in the pretrained large. (e) The output of the training and validation loss of the IndoBERT and attention models in the pretrained base. (f) The output of the training and validation loss of the IndoBERT and attention models in the pretrained large. (g) The output of the training and validation loss of the IndoBERT, BiLSTM, and BiGRU models in the pretrained base. (h) The output of the training and validation loss of the IndoBERT, BiLSTM, and BiGRU models in the pretrained large. (i) The output of the training and validation loss of the IndoBERT, BiGRU, and attention models in the pretrained base. (j) The output of the training and validation loss of the IndoBERT, BiGRU, and attention models in the pretrained large.

the pretrained base and large models. Figures 4(a) and 4(b) show the training and validation loss, and Figures 5(a) and 5(b) depict training and validation accuracy.

4.4.2. The IndoBERT + BiGRU Models. This hybrid model's training and validation loss and accuracy results indicate that the training loss lowers while the validation loss begins to climb after epoch 10. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains constant from epoch 2 to 10 for both the pretrained base and large models. Figures 4(c) and 4(d) depict the training and validation loss, and Figures 5(c) and 5(d) show training and validation accuracy.

4.4.3. The IndoBERT Model + Attention. The hybrid model's training and validation loss and accuracy data indicate that the training loss decreases, but the validation loss grows at epoch 4 on the base model. During the training process, the accuracy of the large model exhibits fluctuations in values until epoch 10. However, the validation loss remains constant from epoch 2 to 10 for both the pretrained and large models. Figures 4(e) and 4(f) show a visual representation of the training and validation loss, and Figures 5(e) and 5(f) depict training and validation accuracy.

4.4.4. The IndoBERT + BiLSTM + BiGRU Models. The hybrid model's training and validation loss and accuracy results indicate a decrease in training loss, but the validation loss remains constant from epochs 4 to 10. During the training process, the accuracy steadily improves until epoch 10. However, the validation loss remains consistent from epoch 3 to 10 for both the pretrained base and big models.

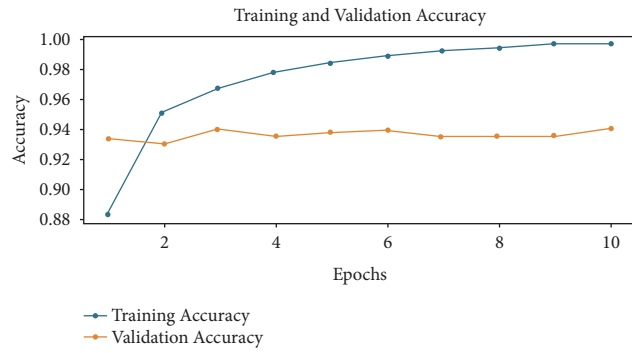
Figures 4(g) and 4(h) depict a graphical representation of the loss, and Figures 5(g) and 5(h) show the accuracy of the training and validation process.

4.4.5. The IndoBERT + BiGRU + Attention Models. The hybrid model's training and validation results indicate a decrease in training loss and an increase in validation loss starting from epoch 2. During the first ten epochs, the training accuracy consistently improves. However, the validation loss remains constant from epoch 1 to 10. This finding remains true for both the pretrained base and large models. Figures 4(i) and 4(j) show the training and validation loss, and Figures 5(i) and 5(j) depict training and validation accuracy.

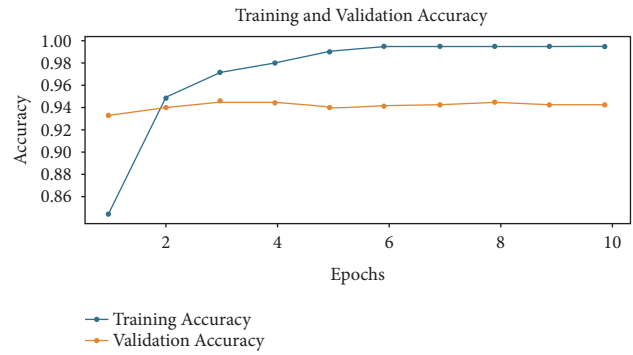
5. Discussion

Exploring different hybrid models integrating IndoBERT with BiLSTM, BiGRU, and attention mechanisms involves adjusting factors such as layer configurations, fine-tuning strategies, and regularization techniques. These adjustments aim to achieve maximum accuracy by enhancing contextual understanding, improving feature representation, effectively handling long-range dependencies, and optimizing focus on essential segments of the input sequence [41].

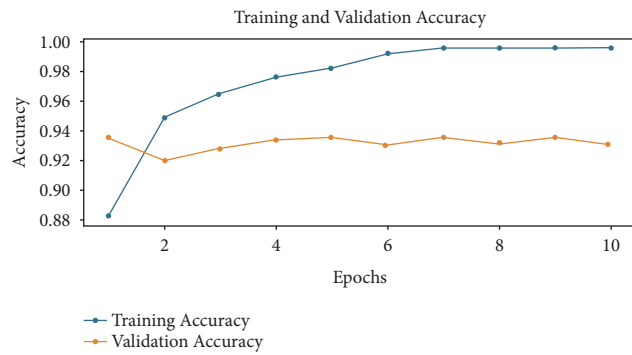
The challenges encountered in processing the Indonesian dataset, including flexible sentence structure and semantic ambiguity, were effectively addressed in the hybrid model design [42]. Sequential models such as BiLSTM and BiGRU were integrated with IndoBERT to capture these flexible syntactic patterns. Contextual embeddings from IndoBERT combined with attention mechanisms help disambiguate these meanings by considering the broader context. These



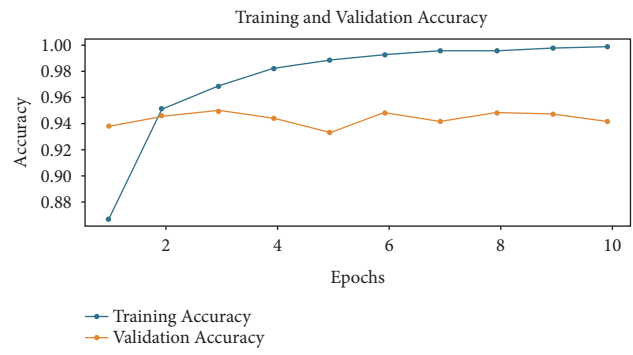
(a)



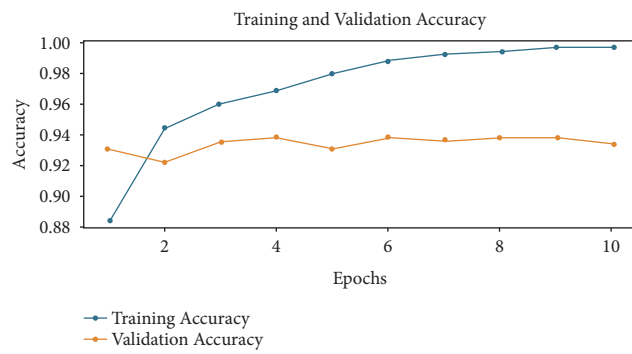
(b)



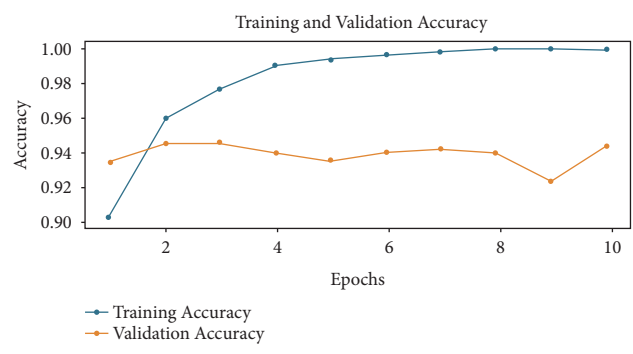
(c)



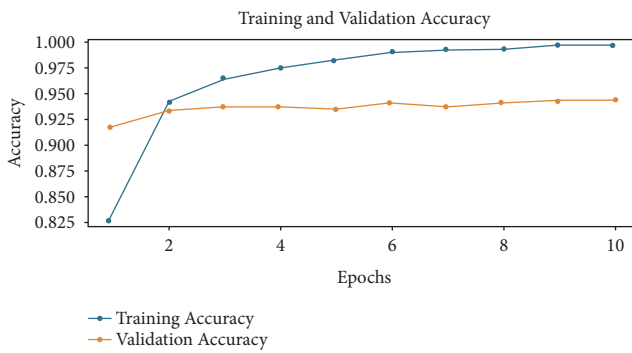
(d)



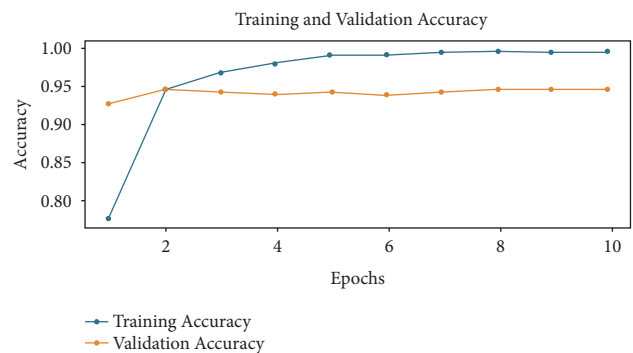
(e)



(f)



(g)



(h)

FIGURE 5: Continued.

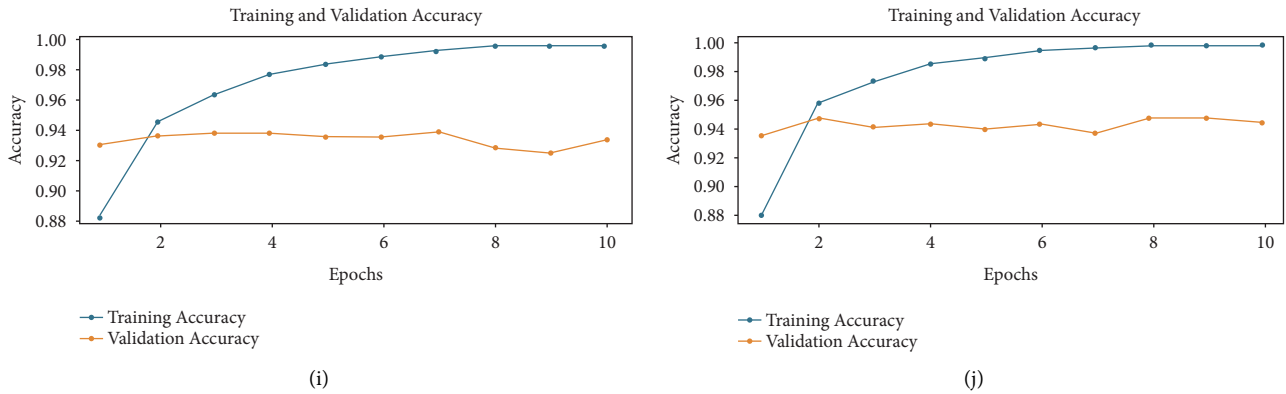


FIGURE 5: The illustration of training and validation accuracy for all models of the sentiment analysis tasks. (a) The output of the training and validation accuracy of the IndoBERT and BiLSTM models in the pretrained base. (b) The output of the training and validation accuracy of the IndoBERT and BiLSTM models in the pretrained large. (c) The output of the training and validation accuracy of the IndoBERT and BiGRU models in the pretrained base. (d) The output of the training and validation accuracy of the IndoBERT and BiGRU models in the pretrained large. (e) The output of the training and validation accuracy of the IndoBERT, BiLSTM, and BiGRU models in the pretrained base. (f) The output of the training and validation accuracy of the IndoBERT, BiLSTM, and BiGRU models in the pretrained large. (g) The output of the training and validation accuracy of the IndoBERT and attention models in the pretrained base. (h) The output of the training and validation accuracy of the IndoBERT and attention models in the pretrained large. (i) The output of the training and validation accuracy of the IndoBERT, BiGRU and attention models in pretrained base. (j) The output of the training and validation accuracy of the IndoBERT, BiGRU and attention models in pretrained large.

embeddings enhance the model's accuracy in understanding and classifying sentiments and emotions [43].

The proposed hybrid model is adaptable to other languages with suitable adjustments and considerations [44]. Although the IndoBERT a robust foundation due to its strong architecture and pretraining on Indonesian text, modifications are essential to accommodate language-specific traits such as morphology, syntax, and semantics.

The improved IndoBERT model is highly versatile. It can be applied to a wide range of natural language understanding tasks beyond sentiment analysis and emotion classification due to its ability to capture contextual information, understand complex linguistic features, and generalize to different NLU tasks [45].

6. Conclusions

The hybrid model, the combination of IndoBERT-modified layers, and the BiLSTM, BiGRU, and attention models were trained using IndoNLU benchmark datasets for emotion classification and sentiment analysis tasks. The results show that combining the IndoBERT-modified layers with the BiLSTM model achieves an accuracy of 93% for sentiment analysis, outperforming the other hybrid models and previous IndoNLU results. Moreover, the hybrid models improve the accuracy of emotion classification. This result is due to the limited training, validation, and testing data in the IndoNLU benchmark dataset. We will train the hybrid

model with other Indonesian emotion classification and sentiment analysis datasets in the future.

A neural network with the IndoBERT model is a promising design for attaining accuracy in NLP tasks. For future work, another variant of the BERT model, such as ALBERT and RoBERTa, could be an alternative design for improving emotion classification and sentiment analysis tasks. Further tuning of the neural network and any variant of the BERT model will also be required to produce significant accuracy in other NLP tasks such as textual entailment and named entity recognition tasks.

Data Availability

The datasets used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The authors declare that they have no conflicts of interest.

Acknowledgments

This work was funded by the Ministry of Education, Culture, Research, and Technology, Directorate for Research, Technology, and Community Services, under the Doctoral Dissertation Research Grant (PDD), contract no. 672/UN11.2.1/PT.01.03/DPRM/2023.

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